

Day 34

Boss Challenge

CSV-report-cli

Goal

Build a command-line application that reads a CSV file and generates useful summary reports from the data.

What will our program do?

User → CSV File → Program reads file
→ Program analyzes data → Program creates report → displays

Input:

CSV file

Process

Read CSV → check if file exists → Load rows → Validate data
→ Calculate data → Calculate statistics → Prepare report

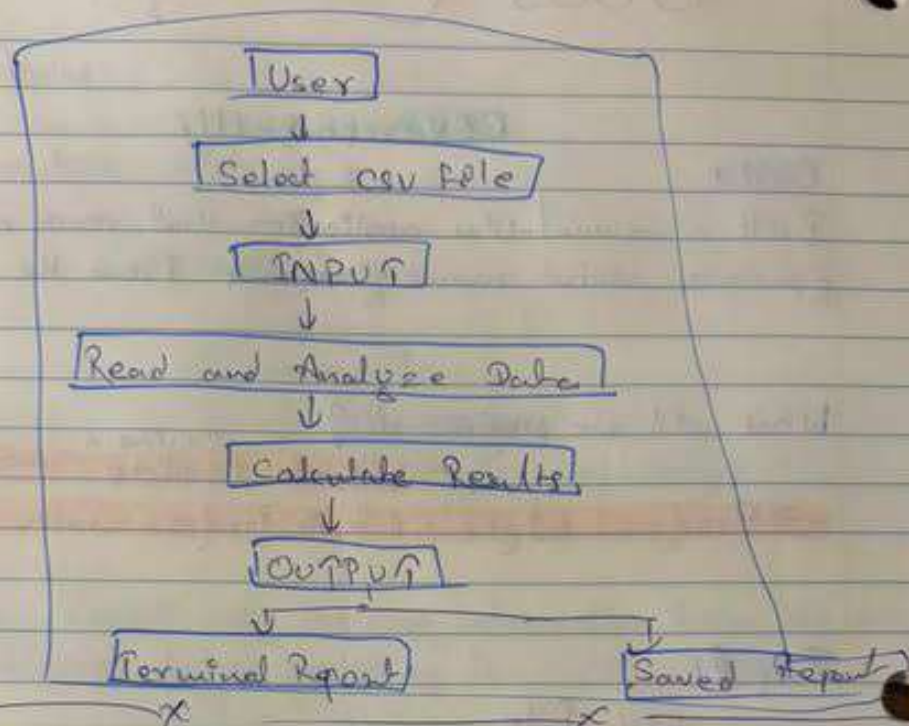
Output

Display in Terminal

----- Report -----

Total Employees : 120
Average Salary : 64000
Highest Salary : 120000
Lowest " : 38000

.txt or .csv file



Project: csv-report-cli

Input

- CSV report
- report type (optional)

Processing

- Read file
- validate "
- Parse CSV
- Analyze data
- Calculate stats
- format report

Output

- Display report in terminal
- Save report to file (optional)

Error handling

Possible error cases

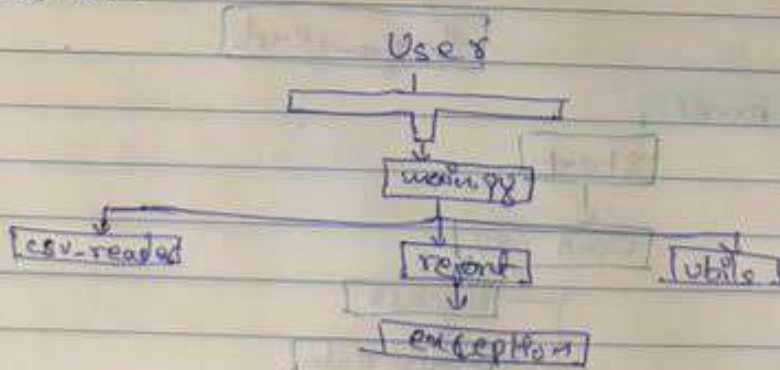
- 1) File not found
- 2) Wrong file type
- 3) Empty CSV
- 4) Missing required columns
- 5) Invalid data
- 6) Permission denied
- 7) Missing command-line argument

Expected behaviour

- Never crash unexpectedly
- Show clear, human-friendly error message
- Explain how the user can fix problem

SRP (Single Responsibility Principle)

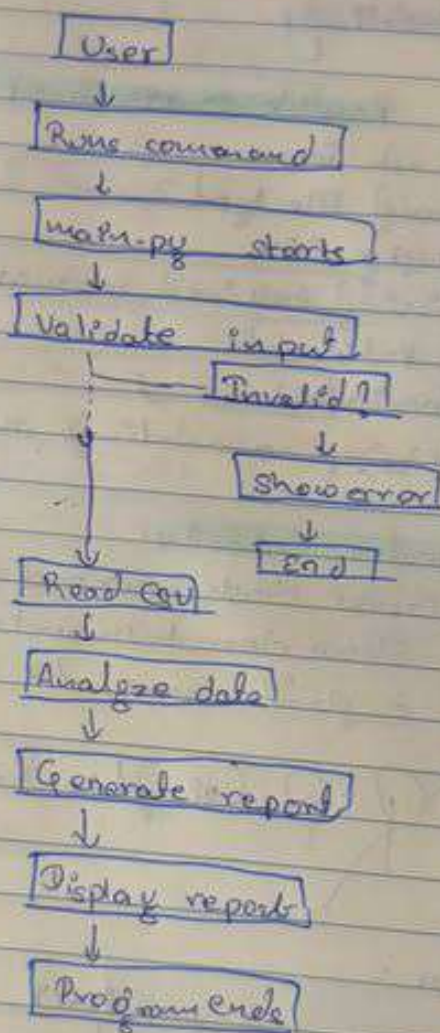
Modules:



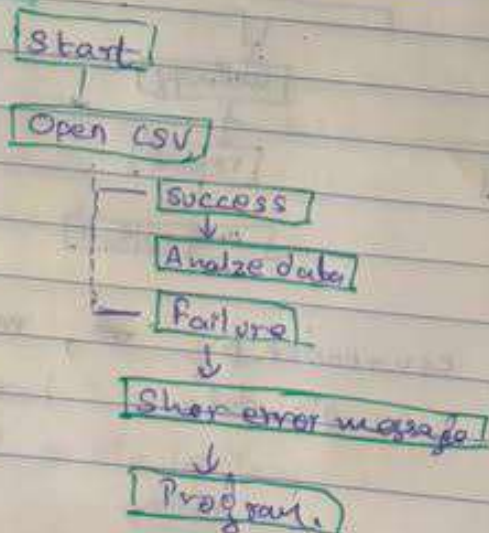
main.py, csv-reader.py, report.py, exceptions.py
utils.py, sample-data, report, documentation

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Program Flow



During error



Program Flow

- 1) ^{Start} program
- 2) Read command-line arguments
- 3) Validate input
- 4) Read CSV file
- 5) Analyze data
- 6) Generate report
- 7) Display report
- 8) End program.

Error Flow

If validation fails:

show error

End program.

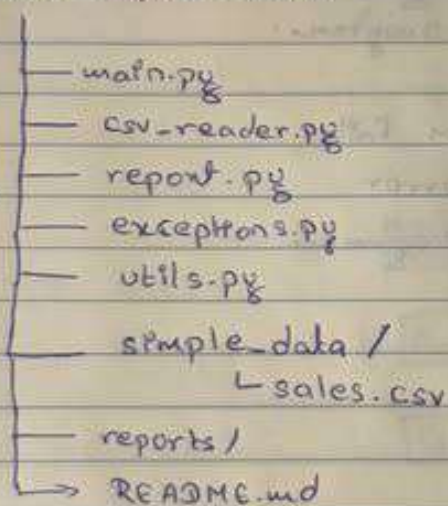
Report

① Project Name
csv-report-cli

② Goal

Read csv files and generate useful summary reports through a command-line interface

③ Structure
csv-report-cli /



④ Module

main.py

starts the app^l and coordinates everything

csv-reader.py

Reads CSV files

report.py

Generates reports from data

exceptions.py

stores custom exceptions

utils.py

Helper functions used throughout the project

⑤

Inputs

- csv filename

- (optional) Report type

⑥ Processing

Read CSV → Validate file → Load data → Analyze data →
Generate report → Format output

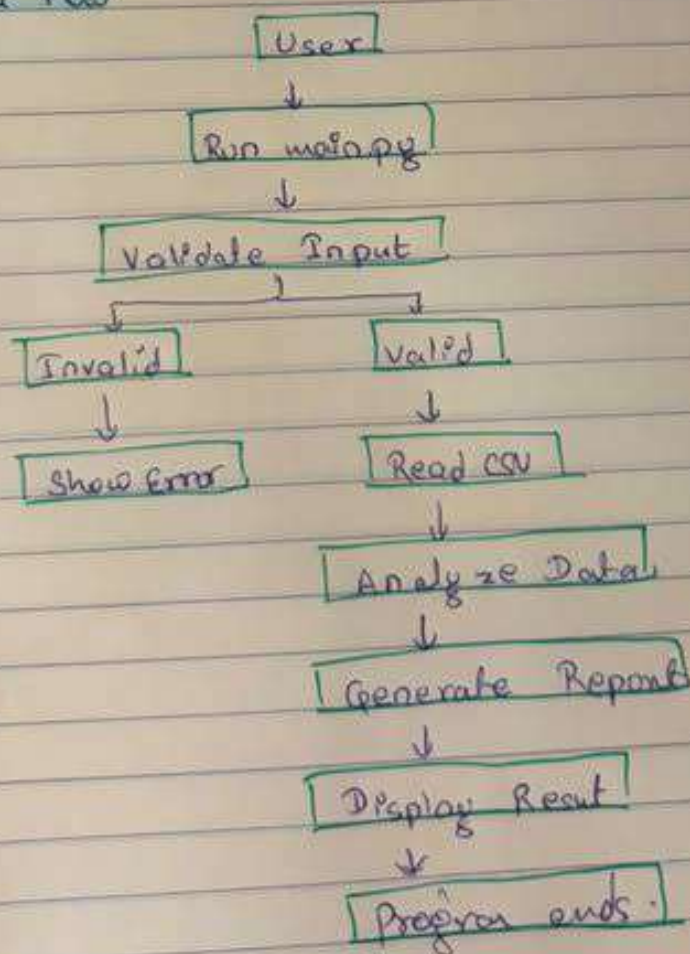
⑦ Output

Terminal report or Saved Report file.

⑧ Error cases

- File not found, - Wrong file type, - Empty CSV.
- Missing columns, - Invalid data, Permission denied,
- Missing command-line argument.

⑨ Program Flow



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